

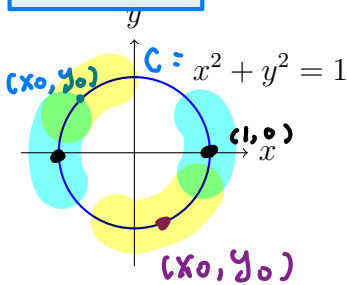
8.2 The Implicit Differentiation

Implicit Functions:

Two variables x and y may be related in an "implicit" way. For example, x and y satisfy the equation $f(x, y) = 0$. Locally, this relation may define y as an implicit function of x or x as an implicit function of y .

Example 2.1

$$x^2 + y^2 = 1$$



$$(x_0, y_0) \in C$$

case 1: If $y_0 > 0$, locally C is

$$y = +\sqrt{1-x^2}$$

case 2: If $y_0 < 0$, locally C is

$$y = -\sqrt{1-x^2}$$

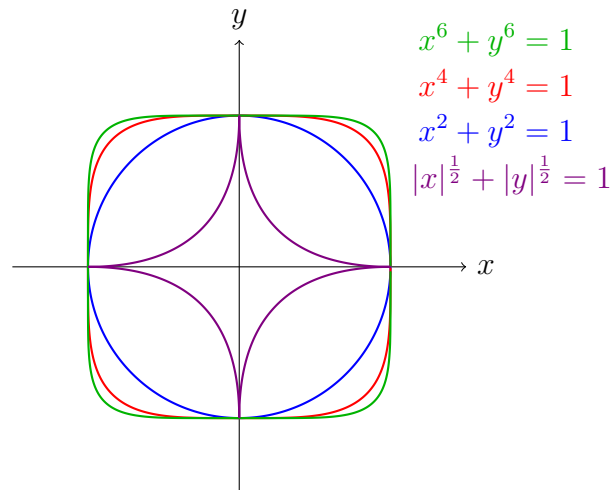
case 3: If $(x_0, y_0) = (\pm 1, 0)$, locally

$$C \text{ is } x = \sqrt{1-y^2} \quad \text{or} \quad x = -\sqrt{1-y^2}$$

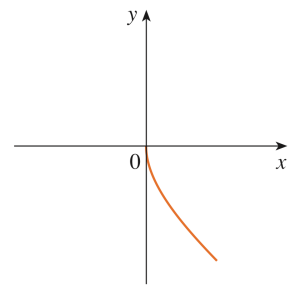
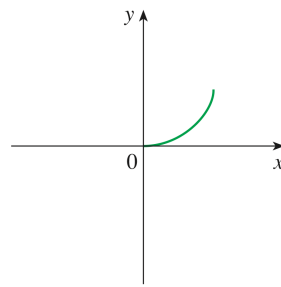
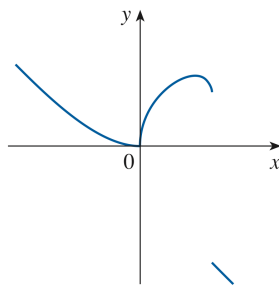
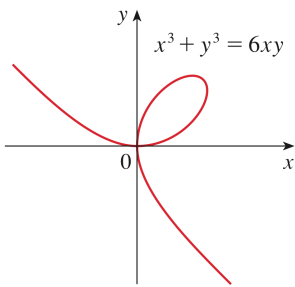
Exercise 2.1

Explore

- (1) $x^{2n} + y^{2n} = 1, n \in \mathbb{N}$.
- (2) $|x|^{\frac{1}{n}} + |y|^{\frac{1}{n}} = 1, n \in \mathbb{N}$.
- (3) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

**Example 2.2**

The folium of Descartes: $x^3 + y^3 = 6xy$.



Compute $\frac{dy}{dx}$ given $f(x, y) = 0$:

- ① Write y as $y(x)$. Differentiate both sides of the equation $f(x, y(x)) = 0$ with respect to x . When you encounter terms that involve y , use the chain rule to calculate the differentiation.
- ② Finally, solve the resulting equation for $\frac{dy}{dx}$.

Example 2.3

For $x^2 + y^2 = 4$, find $\frac{dy}{dx}$ at $(1, \sqrt{3})$.

$$x^2 + (y(x))^2 = 4 \quad \frac{d}{dx} \Rightarrow 2x + 2y \cdot y'(x) = 0$$

$$\text{Hence } 2y \cdot y'(x) = -2x \Rightarrow y'(x) = -\frac{x}{y}$$

$$\text{At } (x, y) = (1, \sqrt{3}), \quad \frac{dy}{dx} = -\frac{1}{\sqrt{3}}.$$

Example 2.4

For $x^3 + y^3 = 6xy$, find the tangent line at $(3, 3)$. At what point in the first quadrant is the tangent line horizontal(vertical)?

sol: Find $\frac{dy}{dx}$ at $(3, 3)$.

$$\textcircled{1} \quad x^3 + (y(x))^3 = 6x \cdot y(x)$$

$$\textcircled{2} \xrightarrow{\frac{d}{dx}} \quad 3x^2 + 3y^2 \cdot y'(x) = 6 \cdot y + 6x \cdot y'(x)$$

$$\text{Hence} \quad (3y^2 - 6x)y' = 6y - 3x^2$$

$$\Rightarrow y' = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$$

$$\text{Find } y' = \frac{2y - x^2}{y^2 - 2x} = 0$$

$$\Rightarrow 2y - x^2 = 0$$

$$\text{Solve } \begin{cases} x^3 + y^3 = 6xy \\ 2y - x^2 = 0 \end{cases}$$

$$\text{At } (3, 3), \quad y' = -1.$$

$$\text{The tangent line at } (3, 3) \text{ is } y = 3 - (x - 3) = 6 - x$$

Example 2.5

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x^4 + y^4 = 1$.

$$\text{sol: } x^4 + (y(x))^4 = 1$$

$$\xrightarrow{\frac{d}{dx}} \quad 4x^3 + 4y^3 \cdot y'(x) = 0$$

$$\therefore \text{e.} \quad x^3 + (y(x))^3 \cdot y'(x) = 0 \quad (1)$$

$$\Rightarrow \text{At } \left(\frac{1}{\sqrt{2}}, \frac{\sqrt[4]{3}}{\sqrt{2}}\right), \quad y'(x) = -\frac{x^3}{y^3} = -\frac{1}{\sqrt[4]{27}}$$

$$\xrightarrow{\frac{d}{dx}} (1) \Rightarrow$$

$$3x^2 + 3y^2 \cdot y' \cdot y' + y^3 \cdot y'' = 0$$

$$\text{At } \left(\frac{1}{\sqrt{2}}, \frac{\sqrt[4]{3}}{\sqrt{2}}\right),$$

$$\frac{3}{2} + \frac{3\sqrt[4]{3}}{2} \cdot \frac{1}{\sqrt[4]{27}} + \frac{\sqrt[4]{27}}{2\sqrt{2}} \cdot y'' = 0$$

$$\Rightarrow y'' = -\frac{4\sqrt{2}}{\sqrt[4]{27}}$$

Example 2.6

Find an equation of the tangent line to the curve $\sqrt{x} + \sqrt{y} = \sqrt{c}$ at (x_0, y_0) where $c > 0$ is a constant. Show that the sum of the x - and y -intercept of any tangent line is equal to c .

Sol: Find $\frac{dy}{dx}$ at (x_0, y_0) .

$$\sqrt{x} + \sqrt{y(x)} = \sqrt{c}.$$

$$\Rightarrow \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \cdot \frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\sqrt{\frac{y}{x}}$$

$$\text{At } (x_0, y_0), \quad \frac{dy}{dx} = -\sqrt{\frac{y_0}{x_0}}.$$

$$\text{The tangent line is } y - y_0 = -\sqrt{\frac{y_0}{x_0}} (x - x_0)$$

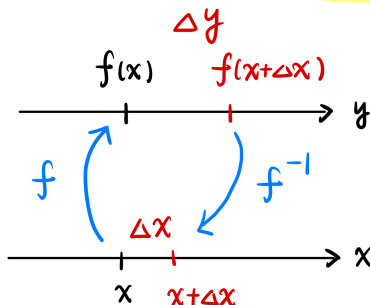
$$\Rightarrow \frac{1}{\sqrt{y_0}} (y - y_0) + \frac{1}{\sqrt{x_0}} (x - x_0) = 0 \Rightarrow \frac{y}{\sqrt{y_0}} + \frac{x}{\sqrt{x_0}} = \sqrt{y_0} + \sqrt{x_0} = \sqrt{c}$$

$$x\text{-intercept: } \sqrt{c} \cdot \sqrt{x_0}, \quad y\text{-intercept: } \sqrt{c} \cdot \sqrt{y_0}$$

$$\sqrt{c} \sqrt{x_0} + \sqrt{c} \sqrt{y_0} = \sqrt{c} (\sqrt{x_0} + \sqrt{y_0}) = \sqrt{c} \cdot \sqrt{c} = c.$$

Derivatives of Inverse Functions
Example 2.7

$y = f^{-1}(x)$ if and only if $f(y) = x$. Find $\frac{dy}{dx}$.



$$f(y(x)) = x$$

$$\Rightarrow f'(y(x)) \cdot y'(x) = 1$$

$$\Rightarrow y'(x) = \frac{1}{f'(y(x))} = \frac{1}{f'(f^{-1}(x))}$$

Example 2.8 $f'(x) = 1 + 2e^{2(x-1)}$, $f'(1) = 3$

$f(x) = x + e^{2(x-1)}$ is one-to-one. Find $\left. \frac{d}{dx} f^{-1} \right|_{x=2}$ and $\left. \frac{d^2}{dx^2} f^{-1} \right|_{x=2}$.

sol: Let $g(x) = f^{-1}(x)$. $f(g(x)) = x$ — (1)

$$\frac{d}{dx} (1) \Rightarrow f'(g(x)) \cdot g'(x) = 1 \quad \text{--- (2)}$$

$$(2) \Rightarrow g'(x) = \frac{1}{f'(g(x))}. \quad \text{Thus, } g'(2) = \frac{1}{f'(g(2))},$$

where $g(2) = f^{-1}(2) = 1$ ($\because f(1) = 2 \therefore f^{-1}(2) = 1$)

$$\text{Hence } g'(2) = \frac{1}{f'(1)} = \frac{1}{3}. \quad (f'(x) = 1 + 2 \cdot e^{2(x-1)})$$

$$\frac{d}{dx} (2) \Rightarrow (f''(g(x)) \cdot g'(x)) \cdot g'(x) + f'(g(x)) \cdot g''(x) = 0$$

$$\text{For } x=2, f''(g(2)) \cdot (g'(2))^2 + f'(g(2)) \cdot g''(2) = 0$$

$$f''(1) \cdot \frac{1}{9} + f'(1) \cdot g''(2) = 0. \quad \text{--- (3)}$$

$$\text{Note that } f''(x) = 4e^{2(x-1)}, \quad f''(1) = 4.$$

$$\text{Thus, } (3) \Rightarrow 4 \cdot \frac{1}{9} + 3 \cdot g''(2) = 0$$

$$\Rightarrow g''(2) = -\frac{4}{27}.$$